Aditya Gambhir

Software Engineer | +1 (951) 544-8605 | gambhir.aditya19@gmail.com | aditya-gambhir-portfolio.vercel.app/ | linkedin.com/in/aditya-gambhir

Education

University of California, Riverside Mar '25

Master of Science | Computational Data Science

Relevant Coursework: Big Data Management, Advanced Operating Systems, Database Systems, Artificial Intelligence, Machine Learning, Spatial Computing, Data Analytics

Vishwakarma Institute of Information Technology May '23

Bachelor of Technology - BTech | Computer Engineering

Relevant Coursework: Data Structures & Algorithms(JavaScript), Database Management Systems, Operating Systems, Computer Networks, Compiler Construction, Computer Architecture, Theory of Computation,

Skills

Programming: Bash · C++ · Java · Javascript · Python · SQL

Backend Technologies: FastAPI · Integration APIs · Microservices · Node.js · REST · Spring Boot

Frontend Technologies: Angular (basic) · CSS · D3.js · HTML · Next.js · React.js · Redux · Tailwind CSS · TypeScript · WebSockets

DevOps & Cloud Infrastructure: Docker · AWS (EC2/S3) · GCP · AWS Lambda · GitHub Actions · Jenkins CI/CD · Kubernetes

Databases: MongoDB · ChromaDB · MySQL · PostgreSQL · SQLite · Redis · TimescaleDB · Database Optimization

Data Engineering & Machine Learning: Hadoop · NumPy · OpenCV · Pandas · Pytorch · Scikit-Learn · Spark · Tensorflow

Development Tools: Agile Methodologies · Cursor IDE · Gemini CLI · Git · Github · Postman · Pytest · Jest · Render · Vercel · VS code

Experience

Tech Mahindra **Pune, India**

Software Development Intern Jun '22 - May '23

- Automated contract intake using JavaScript and a React user interface, cutting processing time by 35%. This involved building React 18 + TypeScript forms on Spring Boot REST, with SQS PDF/email and indexed PostgreSQL.
- Developed a React/TypeScript analytics web application using Recharts and TanStack Table, accelerating sales insights generation by 30%. Leveraged Browser APIs and benchmarking to optimize front-end performance.
- Delivered an internal Mock API Service enabling 500+ developers to simulate 15+ REST endpoints, improving integration testing speed by ~15% using JavaScript, Node.js, Swagger, PostgreSQL+Prisma, Redis rate limits, and API Gateway.
- Fortified authentication/delivery pipelines via OAuth2/OIDC, resolving 5 critical vulnerabilities and slashing PR review time by 20%, enabling faster deployments across microservices with robust security protocols.

C-DAC (Centre for Development of Advanced Computing) **Remote**

Research and Development Intern Jan '22 - Jun '22

- Built a FastAPI + WebSockets service using React and ES6, ingesting P300 data via Lab Streaming Layer and streaming to TypeScript dashboards. Implemented asynchronous programming for efficient data handling, using PostgreSQL, Plotly.js, in-memory buffers, and GitHub Version Control.
- Hardened security to safeguard research data, delivering role-based access and full auditability with 100% JWT/OIDC coverage and 90-day audit retention, using Auth0 (OIDC), RBAC scopes, JSONB audit logs, and TLS via Nginx in Docker.
- Spearheaded cross-functional code reviews with senior researchers and interns, presenting weekly lab demos and driving a 25-30% increase in daily recording blocks through optimized FastAPI, React/TypeScript, Plotly.js, and TimescaleDB integrations.

Projects

Navigate LA28: Real-Time Geospatial Navigation Dec '24

React.js, FastAPI, Spark, Docker, Hadoop

- Engineered a FastAPI-React geospatial platform using HTML, CSS, and ensuring cross-browser compatibility to accelerate map queries by 30%, using Spark/Hadoop distributed processing, PostGIS spatial indexing, and Leaflet.js to enable LA28 venue-scale routing.
- Reduced routing errors by 20% and improved trip accuracy by applying PostGIS spatial joins with GDAL, GeoPandas, and Shapely for multimodal pathfinding, delivering personalized POI recommendations from live crowd data.
- Orchestrated Dockerized microservices on Kubernetes with Nginx, health checks, and OpenAPI/Swagger to achieve 99.99% uptime and fault tolerance, streamlining CI/CD releases and observability for geospatial data processing.

Dockership: Freight Ship Management System Dec '24

Streamlit, MongoDB, Docker, Pytest

- Conceptualized and deployed a containerized freight-ops platform (Streamlit, MongoDB Atlas, Docker) to optimize container allocation and weight balancing, cutting resource misallocation by 20% and operating costs by 12% via real-time 2D visualization and rules-based logic.
- Automated a manifest pipeline; JavaScript services + GraphQL APIs, to remove manual handling, reducing processing time 75% and near-eliminating data-entry errors; enforced quality with schema validation, Pytest unit/integration tests, and audit logging.
- Deployed Spark-on-AWS workflow scheduling to accelerate vessel turnarounds by 25% and guarantee 100% manifest accuracy under noisy input, using idempotent jobs, retry/backoff, checkpointing, and Dockerized task runners with health checks and monitoring.